Potentials and Fields

Problem 10.1

Show that the differential equations for V and $\boldsymbol{A}$

$$\nabla^2 V + \frac{\partial}{\partial t}(\nabla \cdot \boldsymbol{A}) = -\frac{1}{\epsilon_0}\rho$$

$$\left(\nabla^2 \boldsymbol{A} - \mu_0\epsilon_0 \frac{\partial^2 \boldsymbol{A}}{\partial t^2}\right) - \nabla\left(\nabla \cdot \boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) = -\mu_0 \boldsymbol{J}$$

can be written in the more symmetrical form

$$\Box^2 V + \frac{\partial L}{\partial t} = -\frac{1}{\epsilon_0}\rho$$

$$\Box^2 \boldsymbol{A} - \nabla L = -\mu_0 \boldsymbol{J}$$

where

$$\Box^2 \equiv \nabla^2 - \mu_0\epsilon_0\frac{\partial^2}{\partial t^2} \quad \text{and} \quad L \equiv \nabla \cdot \boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}$$

Solution:

Put the definitions of $\Box^2$ and L into the equations for V:

$$\left(\nabla^2 - \mu_0\epsilon_0\frac{\partial^2}{\partial t^2}\right)V + \frac{\partial}{\partial t}\left(\nabla \cdot \boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) =$$

$$= \nabla^2 V - \mu_0\epsilon_0\frac{\partial^2 V}{\partial t^2} + \frac{\partial(\nabla \cdot \boldsymbol{A})}{\partial t} + \mu_0\epsilon_0\frac{\partial^2 V}{\partial t^2} =$$

$$= \nabla^2 V + \frac{\partial}{\partial t}(\nabla \cdot \boldsymbol{A}) = -\frac{1}{\epsilon_0}\rho \; \#$$

and for $\boldsymbol{A}$:

$$\left(\boldsymbol{\nabla}^2 - \mu_0\epsilon_0\frac{\partial^2}{\partial t^2}\right)\boldsymbol{A} - \boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) =$$

$$= \left(\boldsymbol{\nabla}^2\boldsymbol{A} - \mu_0\epsilon_0\frac{\partial^2 \boldsymbol{A}}{\partial t^2}\right) - \boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) = -\mu_0\boldsymbol{J} \ \#$$

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Problem 10.2

Problem 10.3

Problem 10.4

Problem 10.5

Problem 10.6

Problem 10.7

Problem 10.8

Problem 10.9

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Problem 10.37

Problem 10.38